UNITY
GAMEJAM

Unity Crash Course 2026

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Controllo presenze

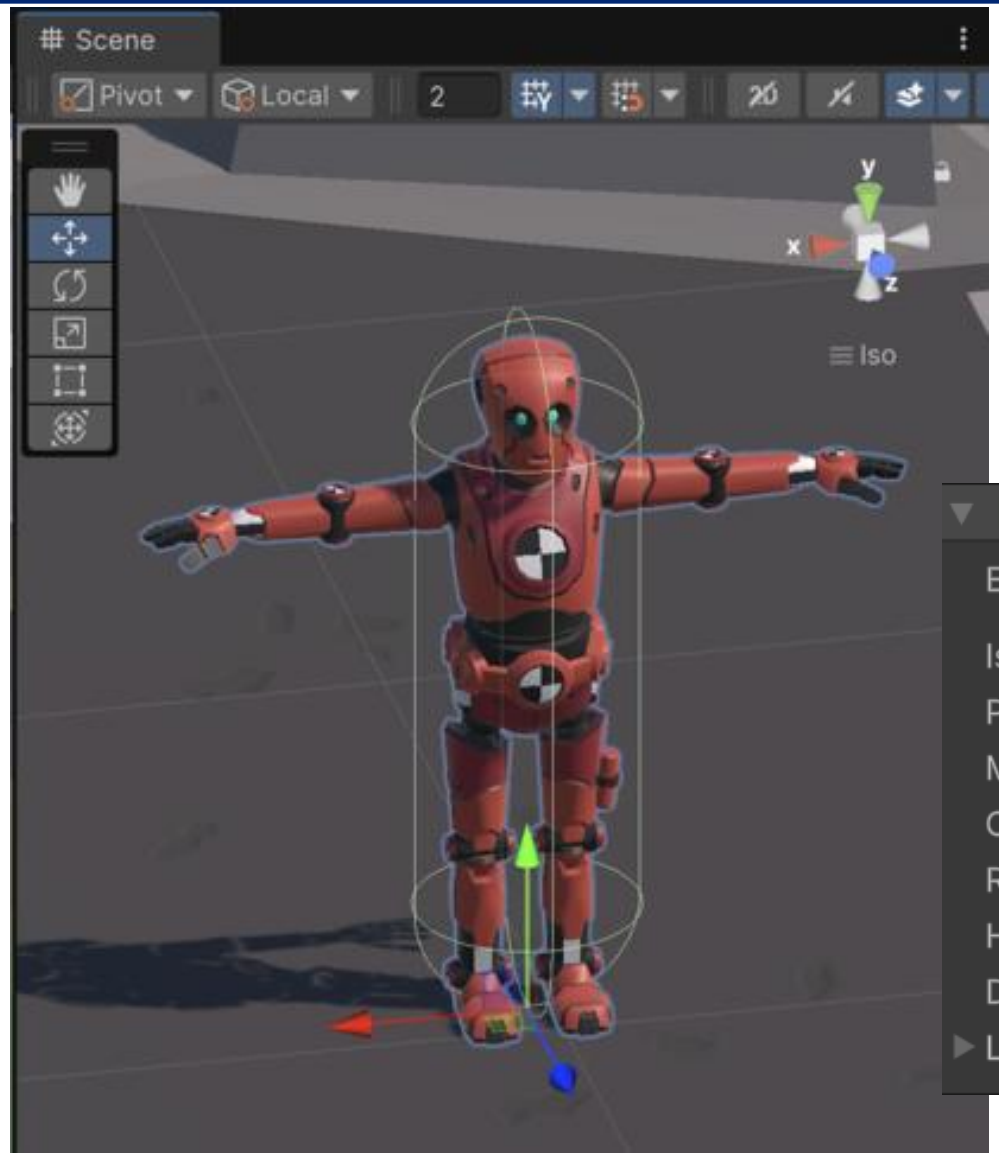
Programma

1. Fisica
2. Movimento avanzato
3. Interazione tra oggetti
4. Prefab
5. Effetti visivi
6. Fondamenti di game logic

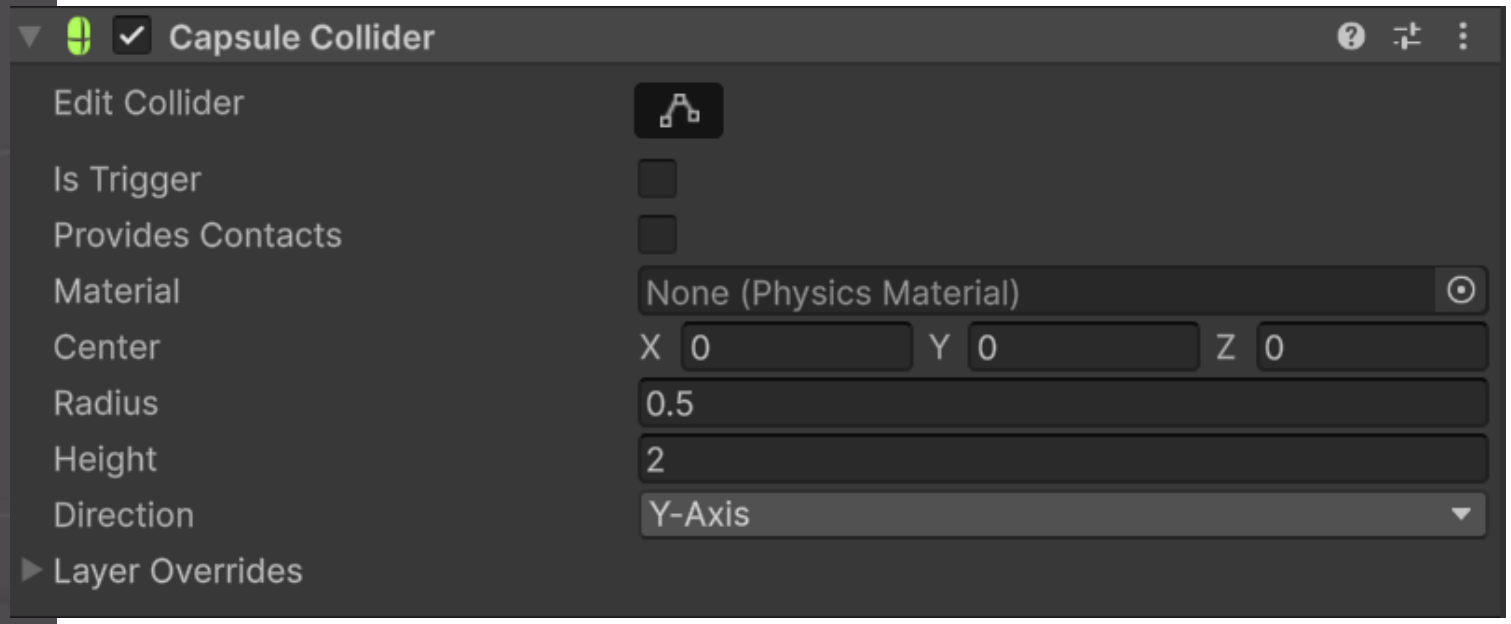
Risorse

uniud-game-jam.netlify.app

Collider

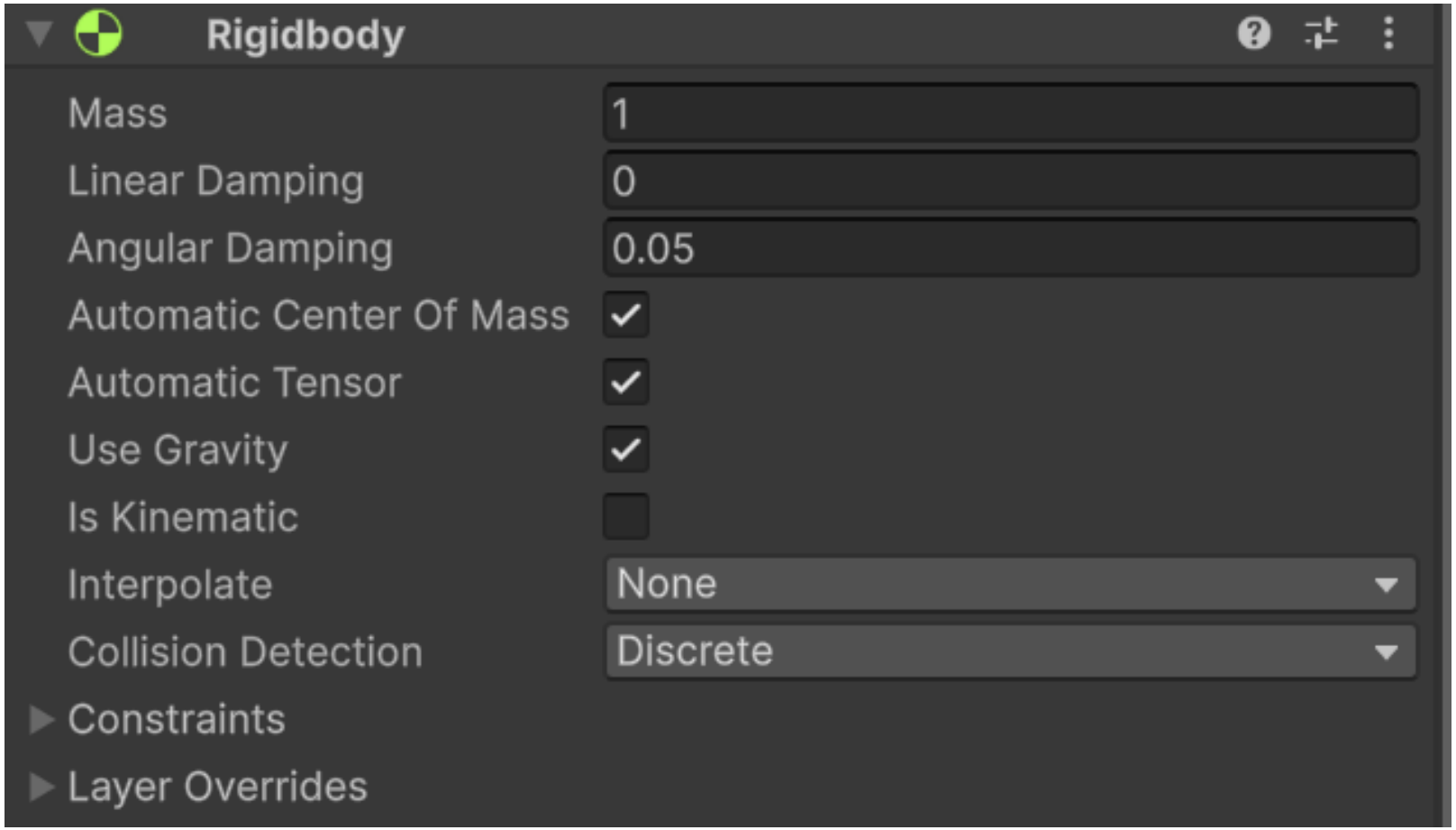


- **Definiscono la forma fisica di un oggetto**, sono indispensabili per gestire collisioni e interazioni con altri oggetti

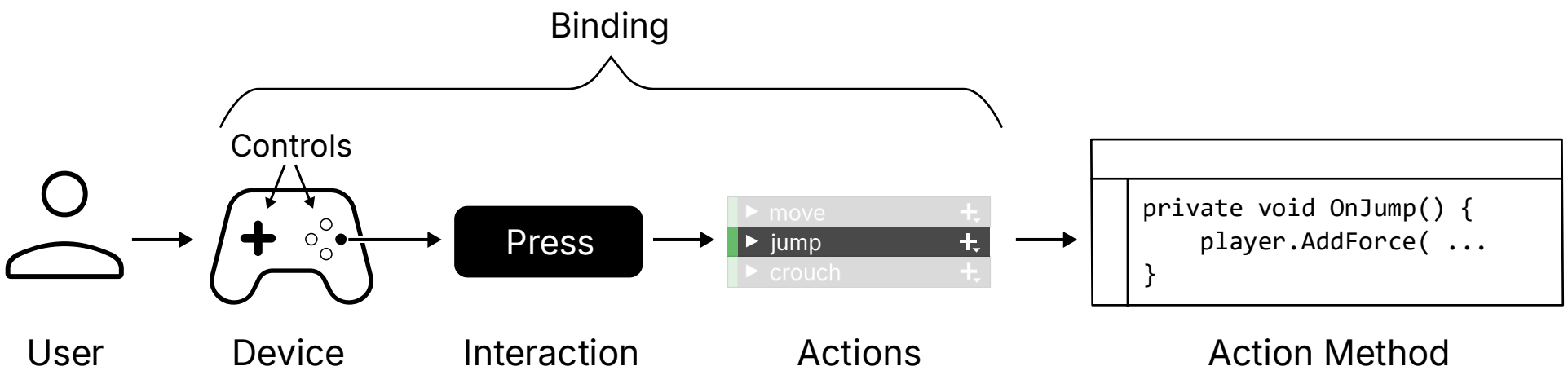


Rigidbody

- Si abilita la fisica sull’oggetto



Input System



Movimento Semplice

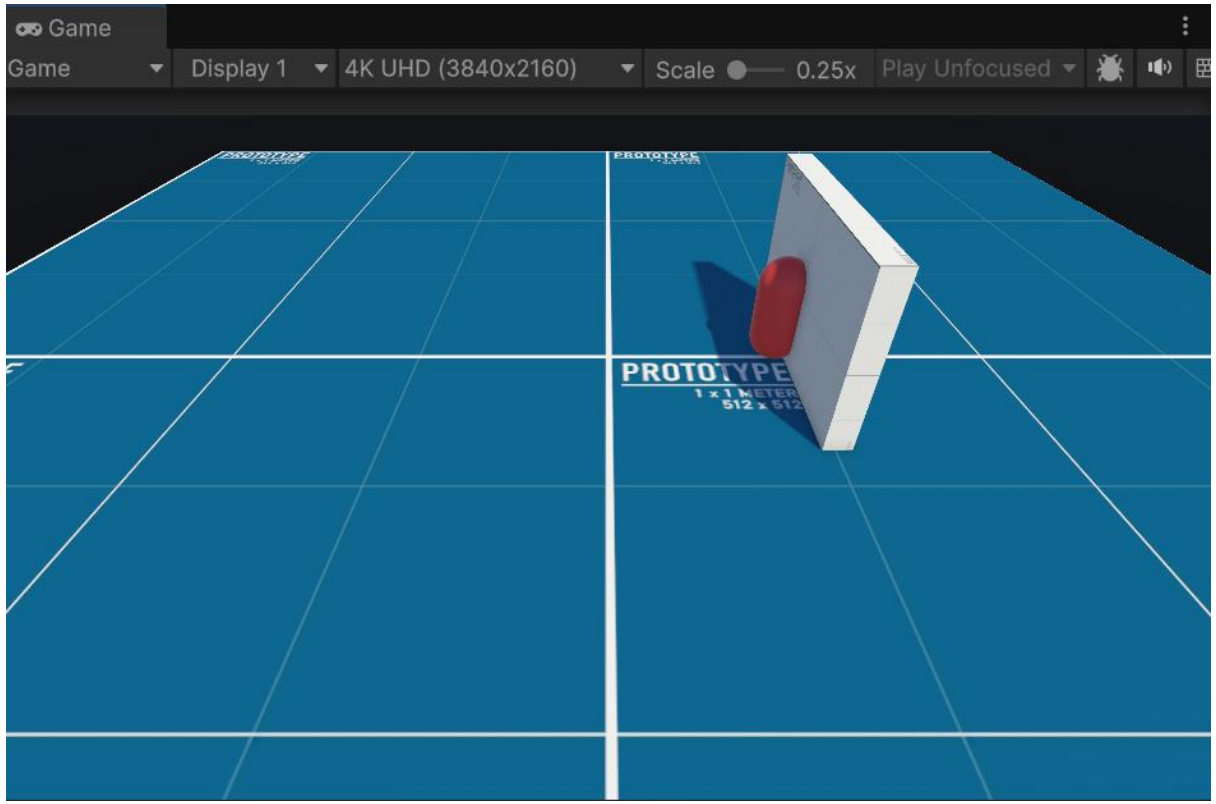
root/scripts/PlayerControllerSimple.cs

Problema

Non permette alcuna interazione con l'ambiente, ad esempio con degli ostacoli.

Soluzione

Character Controller o Rigidbody



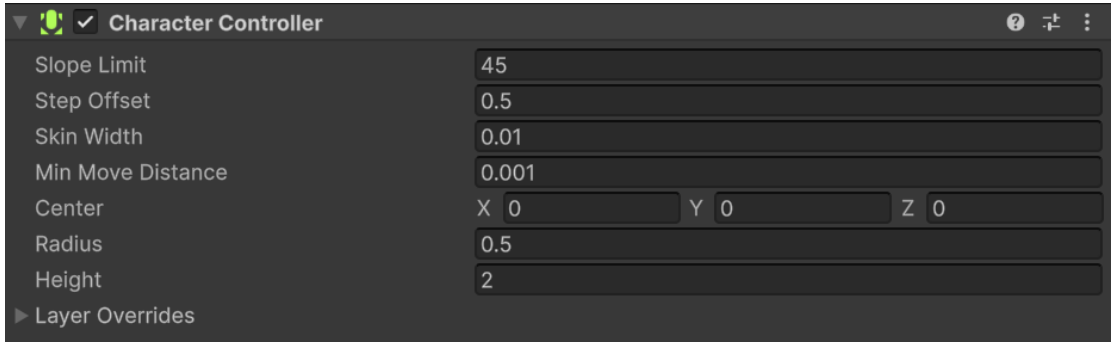
Movimento Avanzato

Character Controller

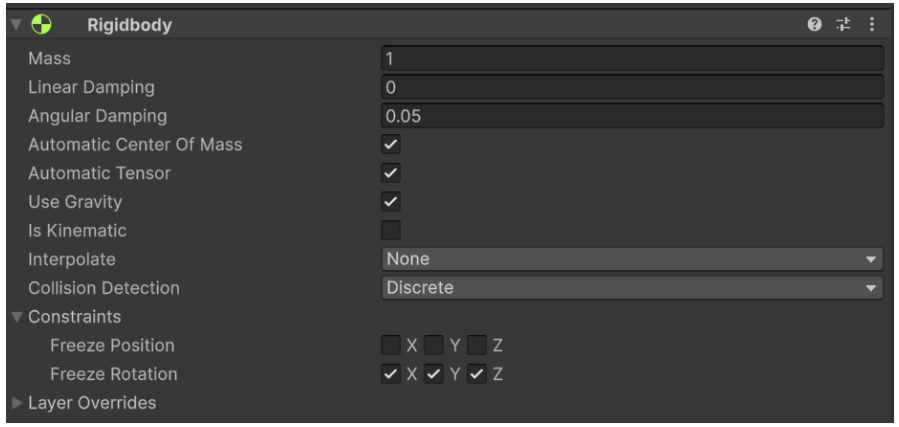
vs.

Rigidbody Controller

- Sufficiente nella quasi totalità dei casi. Molti scenari già implementati (scale, rampe, ...)
- **Più semplice** da usare per il movimento del personaggio.
- Rileva le collisioni, ma non applica alcuna forza fisica sugli oggetti.

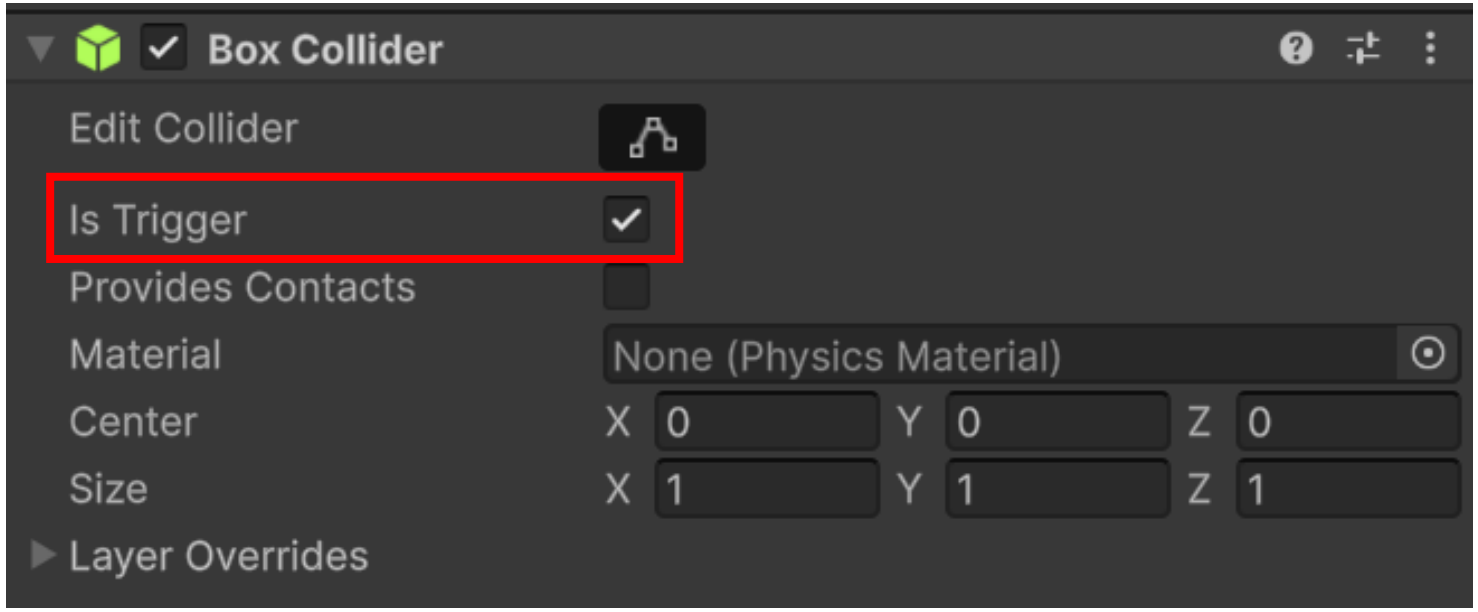


- Utile **solo se** interazioni fisiche con l'ambiente sono **indispensabili**.
- Codice **più complesso**, necessario gestire forze fisiche a mano.
- Si basa sulle leggi della fisica e interagisce con altre entità fisiche tramite collisioni e forze.



Trigger

- Disabilita le collisioni.
- Utile per attivare dei comportamenti quando il giocatore entra in contatto con degli oggetti.



Trigger

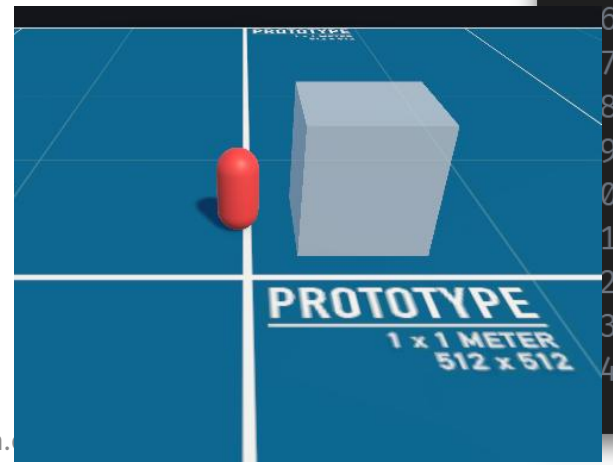
Codice: Trigger 1.cs

```
using UnityEngine;

public class Trigger : MonoBehaviour {

    private void OnTriggerEnter(Collider other) {
        Debug.Log("Entered trigger" + other.name);
    }

    private void OnTriggerExit(Collider other) {
        Debug.Log("Exited trigger " + other.name);
    }

    private void OnTriggerStay(Collider other) {
        // Debug.Log("Staying in trigger");
    }
}

```

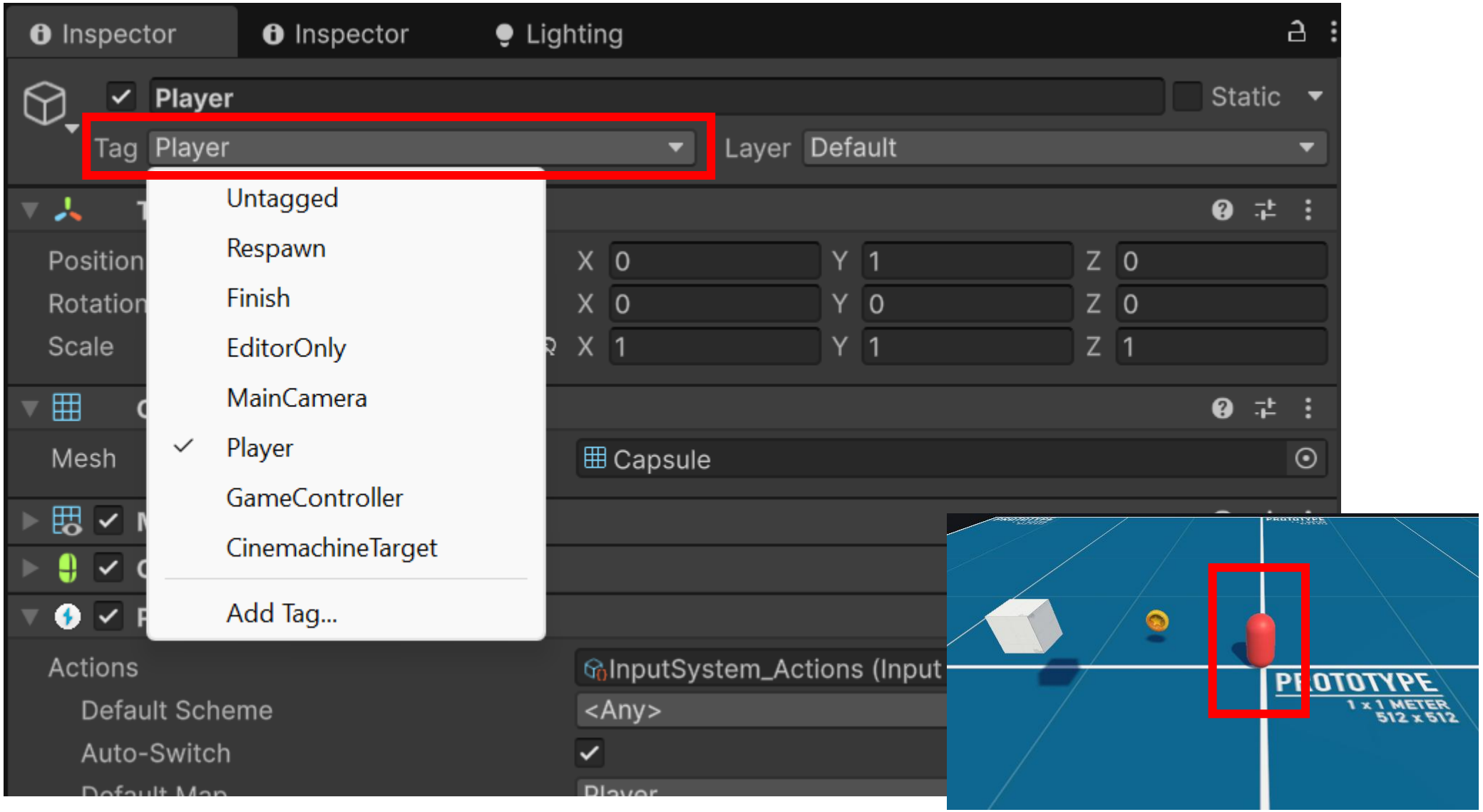
Trigger 2

Codice: Trigger 2.cs

```
using UnityEngine;

public class Trigger : MonoBehaviour {

    public Material materialEnter;
    public Material materialExit;
    private Renderer objRenderer;

    private void Start() {
        objRenderer = GetComponent<Renderer>();
        objRenderer.material = materialExit;
    }

    private void OnTriggerEnter(Collider other) {
        Debug.Log("Entered trigger: " + other.name);
        objRenderer.material = materialEnter;
    }

    private void OnTriggerExit(Collider other) {
        Debug.Log("Exited trigger: " + other.name);
        objRenderer.material = materialExit;
    }
}
```



Tag


Inspector
Inspector
Lighting
Player
Static
Tag
Player
Layer
Default
Untagged
Respawn
Finish
EditorOnly
MainCamera
Player
GameController
CinemachineTarget
Add Tag...
Position
X 0
Y 1
Z 0
Rotation
X 0
Y 0
Z 0
Scale
X 1
Y 1
Z 1
Mesh
Capsule
Actions
InputSystem_Actions (Input
Default Scheme
<Any>
Auto-Switch
PROTOTYPE
1 x 1 METER
512 x 512

Esercizio – Trigger

1 Aggiungere l'acqua (**game object + materiale + script**).

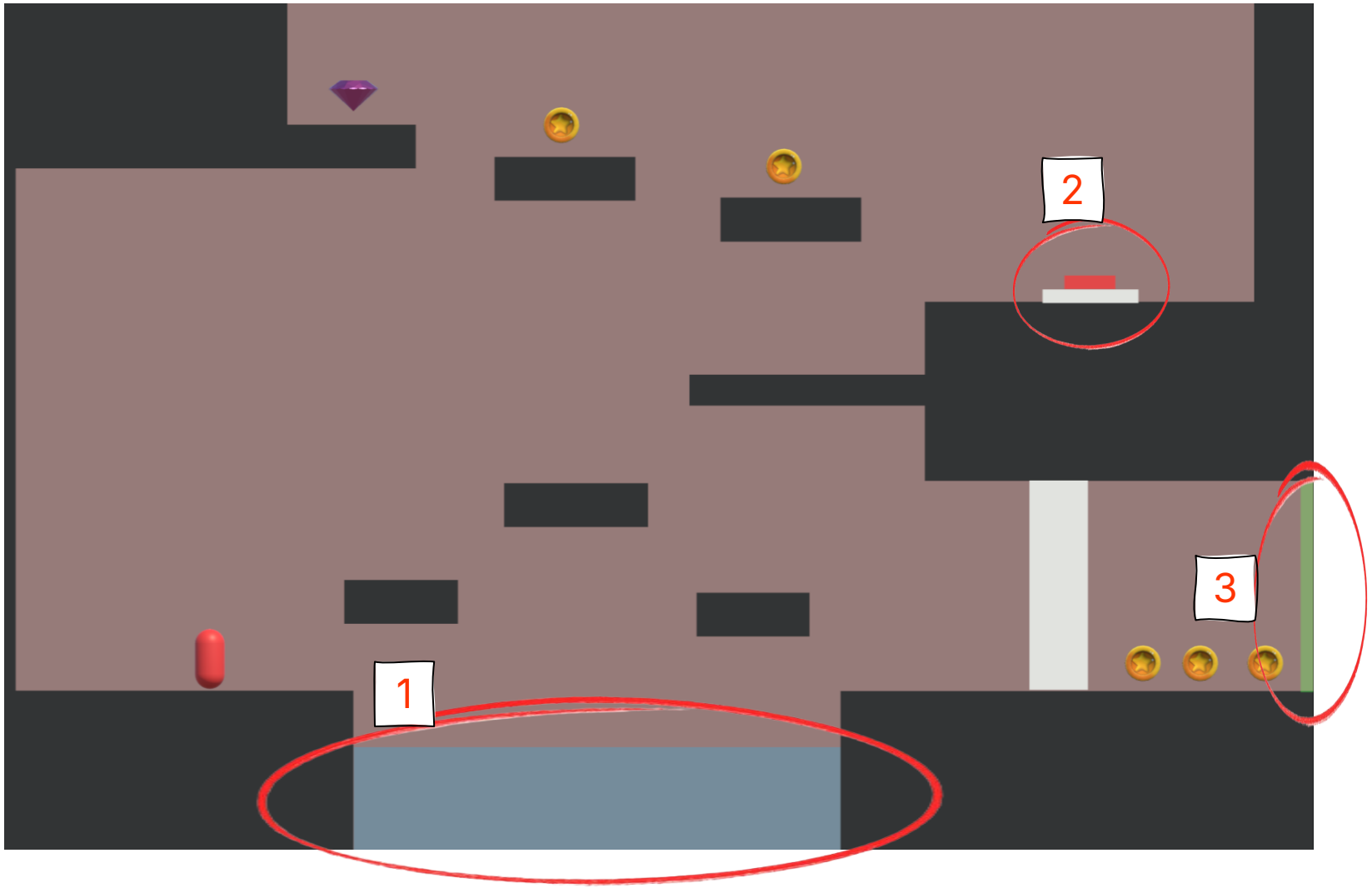
- Se il player entra nell'acqua: `Debug.Log("Game Over")`

2 Aggiungere il pulsante.

- Se il player ci va sopra, la porta scompare `Destroy(porta)`

3 Aggiungere l'area per fine livello.

`Debug.Log("Next Level")`



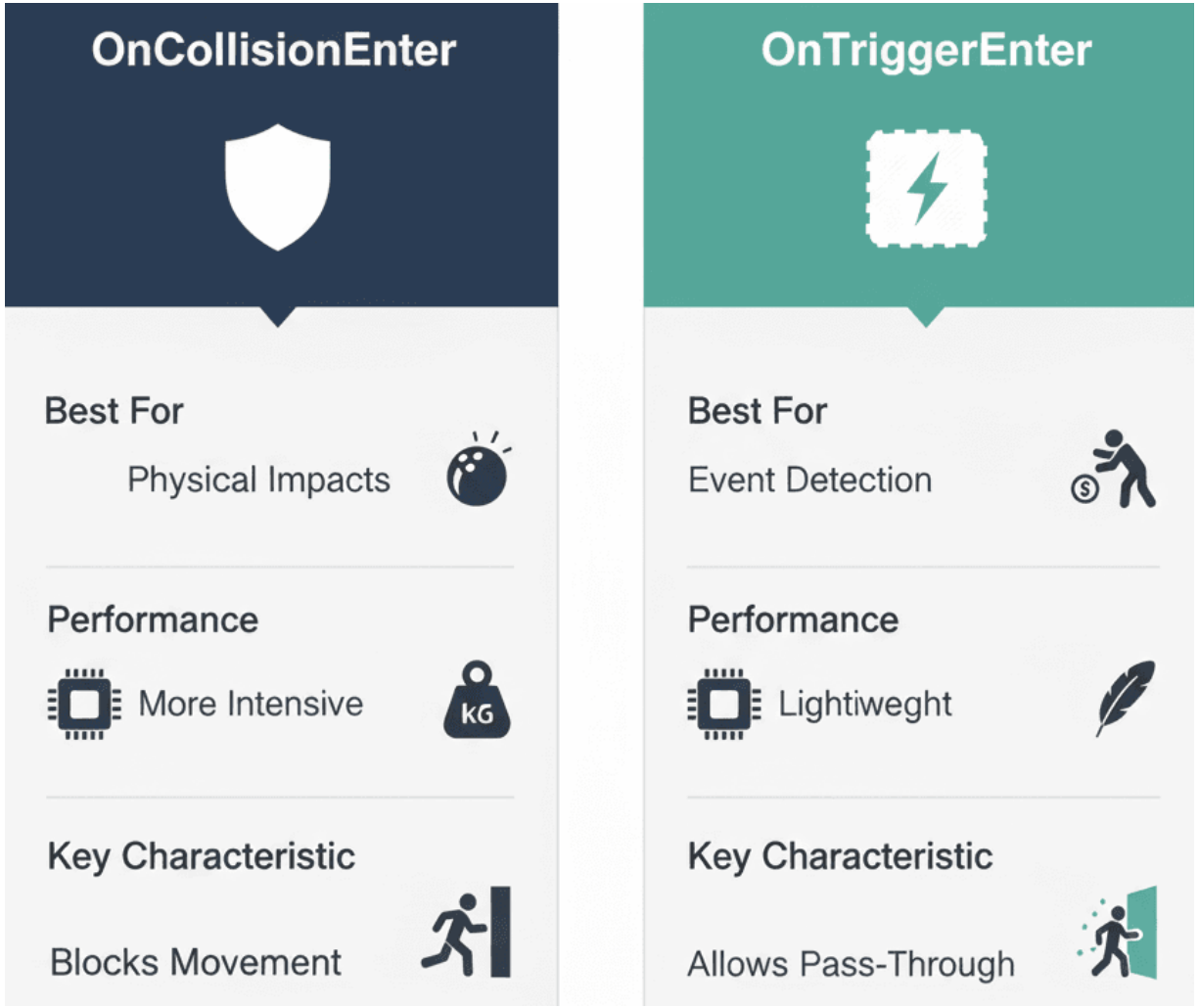
Collider vs Trigger

Collider

```
void OnCollisionEnter(Collision collision) {
    if(collision.gameObject.tag == "Wall") {
        Debug.Log("Hit a wall!");
    }
}
```

Trigger

```
void OnTriggerEnter(Collider other) {
    if(other.CompareTag("Player")) {
        Debug.Log("Player entered the zone!");
    }
}
```



Esercizio – Trigger e Collider

Obiettivo: Salire sulle scale, attivare il pulsante per far cadere il peso sulla scatola, romperla e raccogliere il diamante.

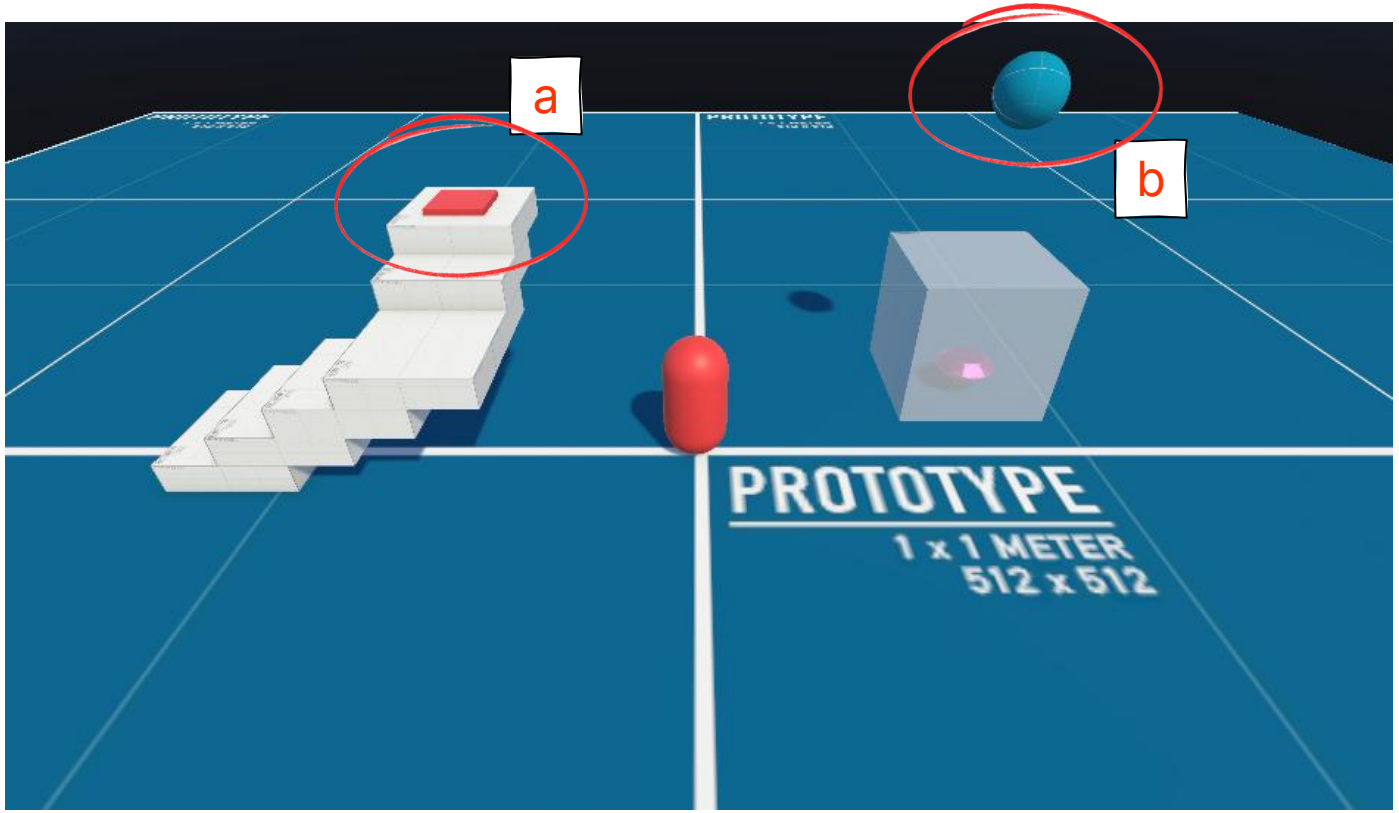
Suggerimento: Per il pulsante che è un trigger usa **OnTriggerEnter**, mentre per il peso che ha un collider, usa **OnCollisionEnter**

1. Ricostruite la scena
2. Scrivete gli script necessari

   a) EnableGravity (onTriggerEnter).
   `pesoRigidbody.useGravity = true;`

   b) DestoryBox (onCollisionEnter)

   Suggerimento : aggiungere un tag alla scatola.



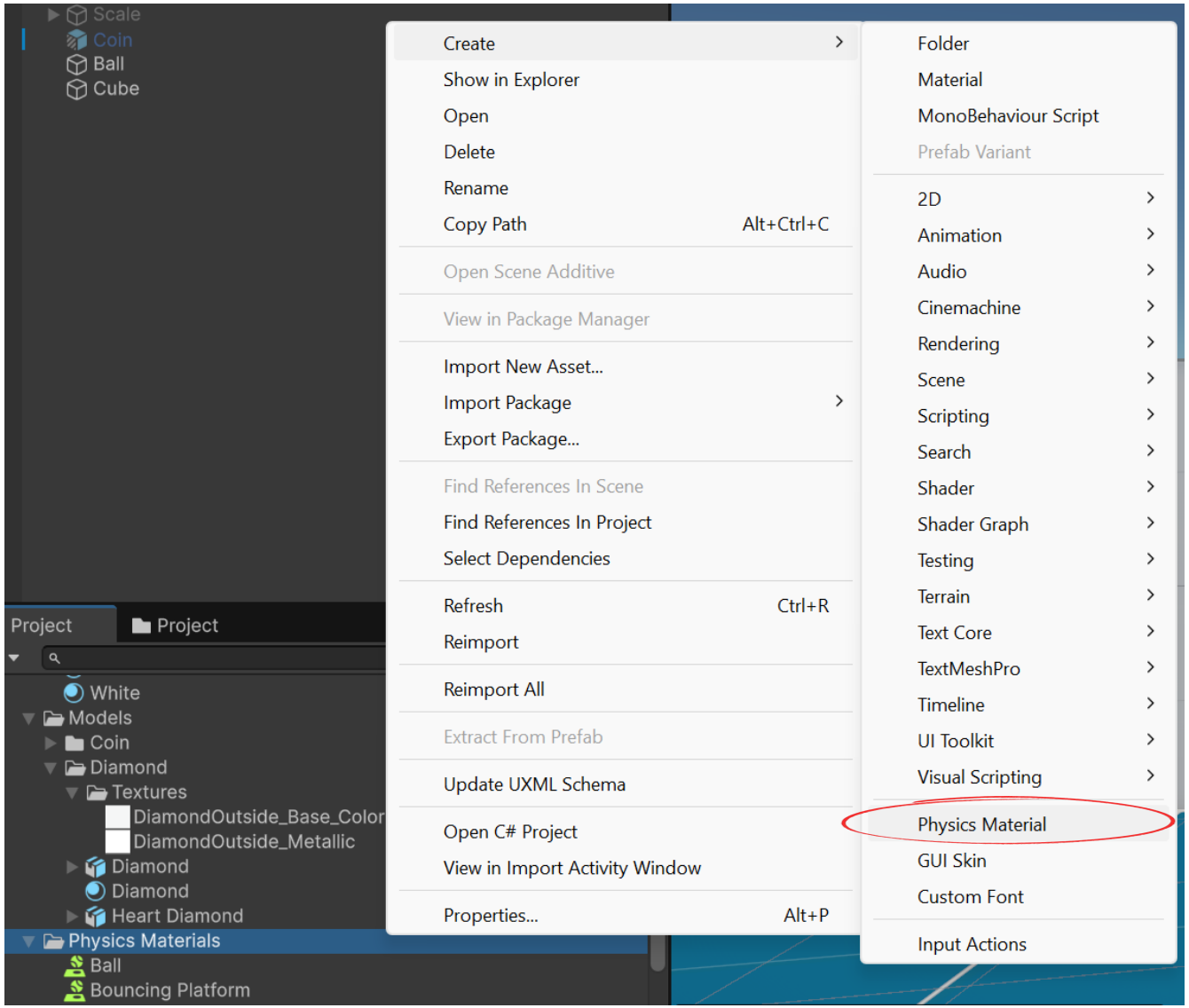
Effetti Particellari

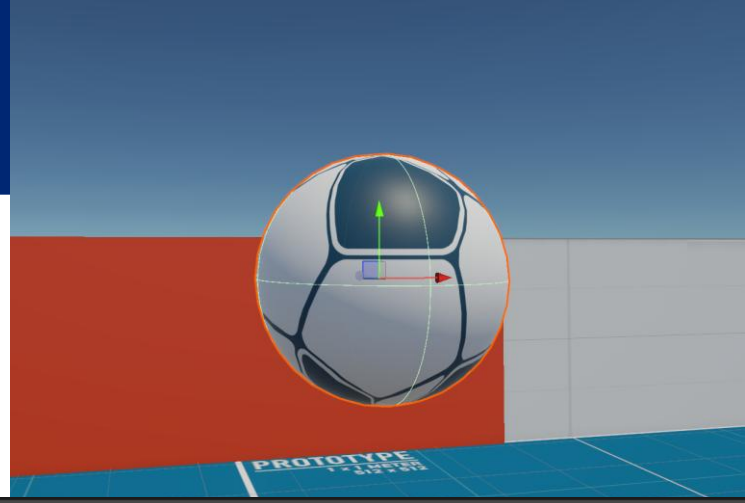
https://assetstore.unity.com/search#q=explosion&nf-ec_price_filter=0...0

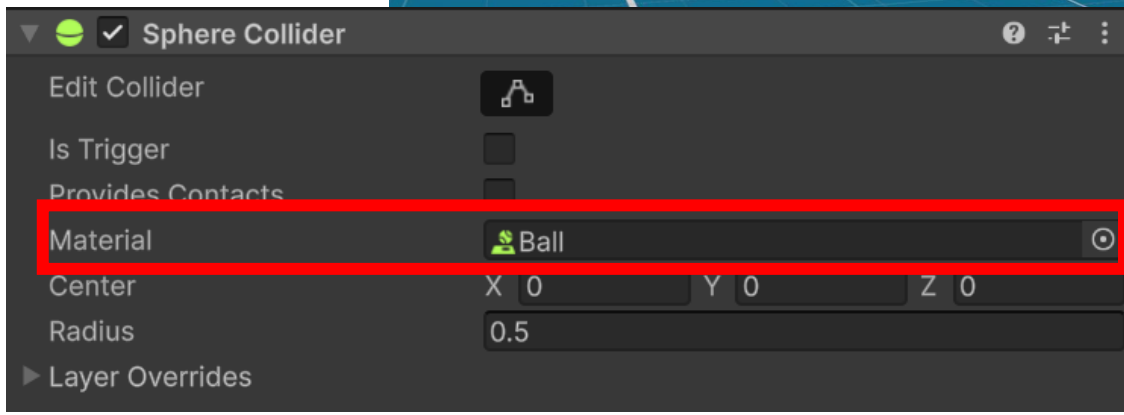
https://assetstore.unity.com/packages/vfx/particles/simple-fx-cartoon-particles-67834

https://assetstore.unity.com/packages/vfx/particles/particle-pack-127325

Materiali Fisici

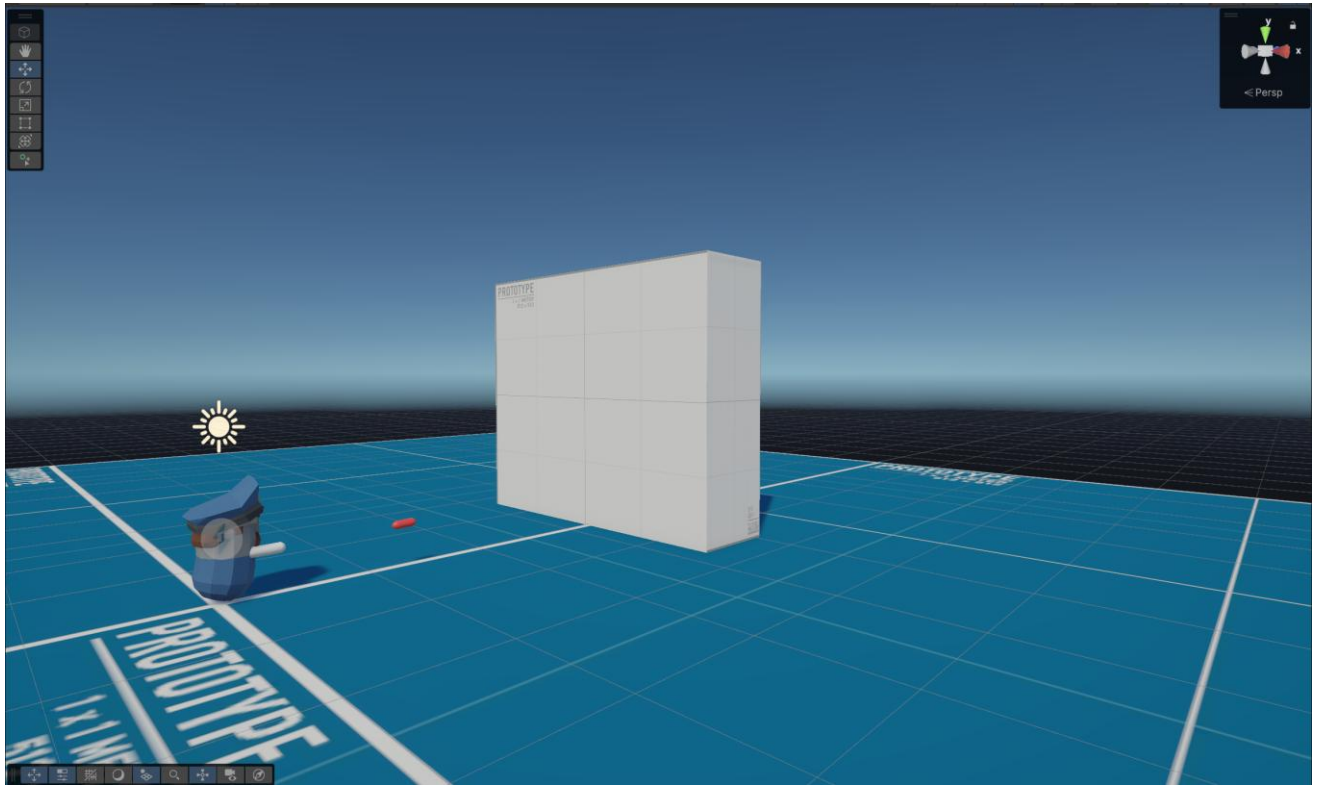




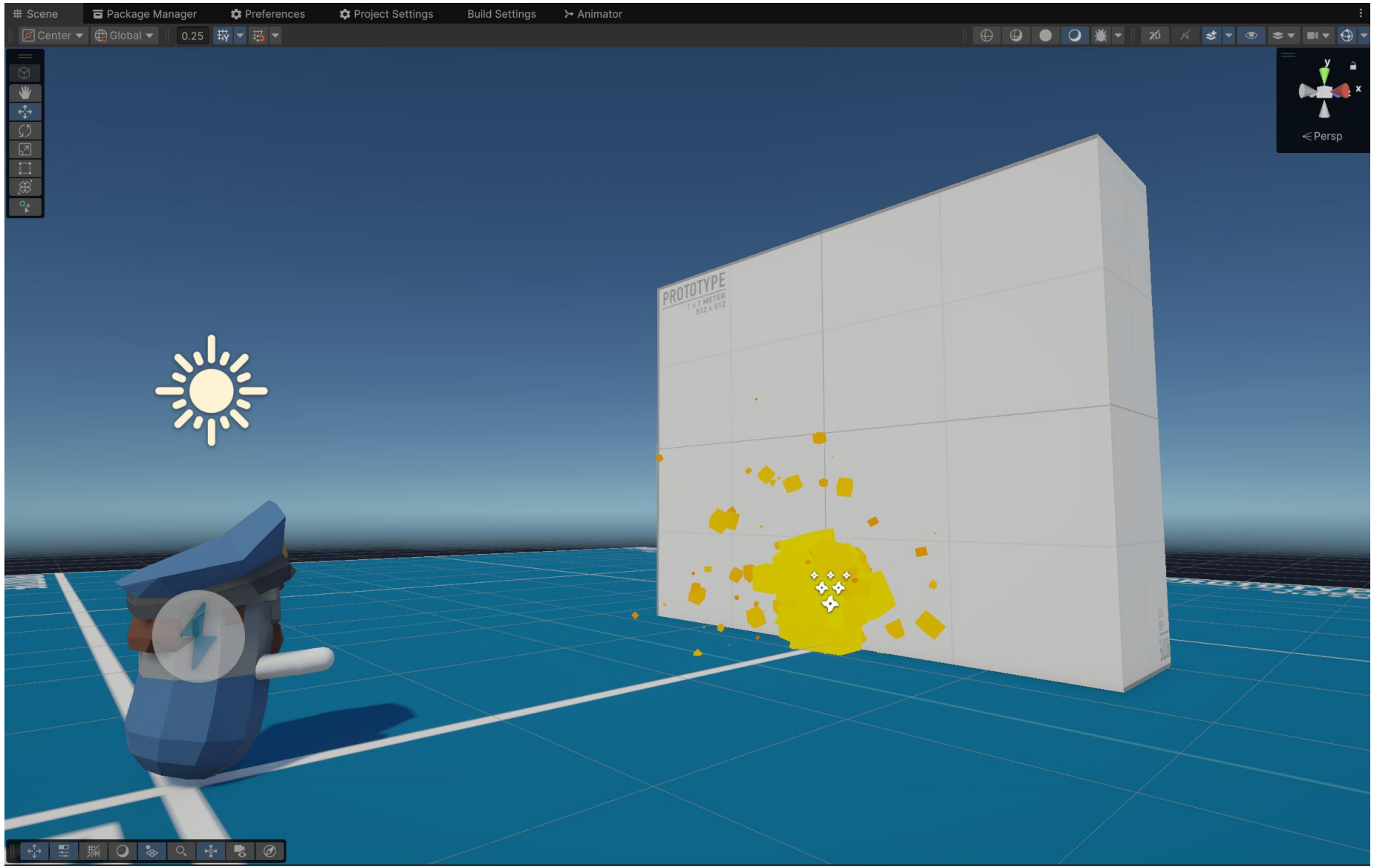


Sparare

1. Prefab per il proiettile con Rigidbody.
2. Creazione del punto di sparo.
3. Script per:
    1. Istanziare il proiettile.
    2. Applicare una forza al proiettile
    3. Effetto particellare quando il proiettile collide con un’altro oggetto



Effetti Particellari



Nella prossima lezione

Giovedì 9 aprile
14:00 – 18:00

1. Interfacce grafiche
2. Audio
3. Animazioni
4. Movimento autonomo dei personaggi

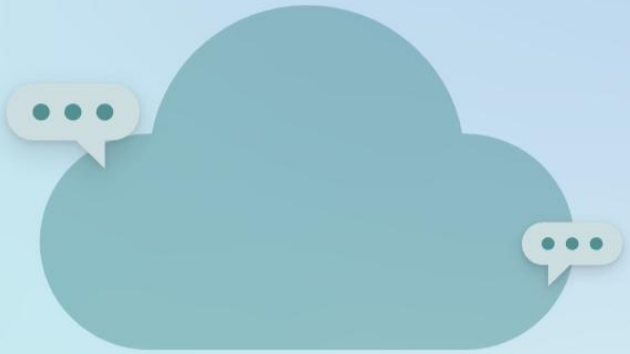
20 responses submitted

Inserisci il tuo codice utente

Scan the QR or use link to join

https://forms.office.com/e/sbuwJQMqsm

Copy link



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud | All responses

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Controllo presenze – fine lezione